

Loop quantum gravity, string theory, whatever you name, none of them have been able to address the cosmological constant problem. There is something very big we are yet to find. In 1933, Wolfgang Pauli discovered that quantum vacuum fluctuations should have stopped the universe from expanding. He wrote that “it’s more consistent to exclude zero-point energy because, evidently from experience, it does not interact with the gravitational field.” We still don’t know why. “We need to go back to square one.” Professor Subir Sarkar of Oxford tells me that “this unsolved problem should have made us suspicious” when astronomers claimed to detect a cosmological constant from supernova data in 1998, something which resulted in the Nobel Prize in 2011. My name’s Curt Jaimungal, and as usual here on Theories of Everything, this podcast is technical because I want to show you the details, since the PopSci accounts are egregiously misleading. To make sense of Sarkar’s claims, it’s useful to know the general story. Here it is. Supernovae are those elephantine star explosions, and certain distant supernovae appear 30% or so fainter than we traditionally expected. The interpretation is that cosmic acceleration is driven by dark energy. This sounds reasonable since if they are fainter than expected, then they are farther than expected, and if they are farther than expected at a given redshift, it means space expanded more than predicted, which implies the expansion rate’s been increasing across time. Increasing expansion rate is the same as acceleration, but this whole interpretation assumes that the universe is perfectly the same no matter how you translate yourself across space or what angle you view it from. This was an assumption made in 1922 when we had almost no data. The FLRW metric is the sine qua non of every dark energy inference. Sarkar’s group tested it. The cosmic microwave background, which is that afterglow of radiation from the early universe, shows a hot spot. So one direction appears slightly warmer, potentially because we’re moving toward it. If our motion is what’s causing this hotspot, then distant matter should show the same pattern. But it doesn’t. The matter dipole is twice as large, and this is confirmed at over 5 sigma, meaning it’s a roughly 1 in 3.5 million chance that it’s a fluke. Moreover, the acceleration is directional, not isotropic, as dark energy requires. Today we cover the vagaries of supernova standardization, how cosmologists stratify parameters while violating sacrosanct principles, and why the professor argues that a century-old metric requires a theoretical revolution before acceptance. Professor, I’m excited to be speaking with you. I’ve been prepping for this interview in many respects for years, going through your work and the responses to your work fairly extensively. So welcome, and thank you. Thank you. It’s great to be on. And in fact, I should say I’ve been watching some of the videos you have already recorded. And I was quite struck by, of course, I looked at the ones which caught my interest, like your video with Cumrun Vafa and with John Donoghue and things concerning gravity and cosmology and so on. And I must say, I was impressed that you had done your homework. You ask them very relevant questions and it’s clear that you have a maths background, otherwise you wouldn’t have known what they were talking about. So all that was good. But what struck me was that these, you know, they are of course professionals—that is to say, we make a living out of this—whereas you make a living out of asking us what we do for a living. So that’s a kind of interesting perspective. And you know, one day I think I would like to turn the tables and interview you to ask you what you make out of the fact

that people can have such different viewpoints on what is essentially fundamentally the same, you know, ontologically or philosophically the same issue. And yet we come at it from so many different points of view. But there is supposed to be one underlying truth there and that's the one we are trying to capture. And somehow this whole enterprise works. I mean, we do make progress. I know that there is a school of thought that holds that everything is constructed, but we know bloody well that physical laws aren't constructed. They exist independent of us. And it's kind of quite fascinating to see how we do things without actually being aware of what we are doing. You know, we don't reflect on what we are doing, which philosophers usually do. And so it's kind of very interesting to be someone, I think, like yourself at the boundary between philosophy and physics, as it were, and to ask these questions. That's extremely kind of you, sir. So I have great respect for you as well. And in large part, I'd like this podcast also to be about what sigma results mean. And there are at least two kinds of sigma. So one is that when someone says this result has five sigma, most people don't know that there's a hidden asterisk. Most of the lay public don't know that that just means conditional on the systematic errors not being there, that everything is correctly modeled. So let's call that a nominal sigma. That's the one that's reported on Wikipedia. But then there's an effective sigma, like the actual reliability. Maybe it's systematically lower. Six sigma results have disintegrated because of loose cables or contamination of some sort. These sigma results actually tend to disappear more often than just a straight looking at the sigma, the nominal sigma, would lead you to believe. Yes. So this episode is about dark energy. And I'm going to open with a provocative question. Do you think the 2011 Nobel Prize was prematurely awarded? Well, that's a loaded question. Let me say that it's not really for me to say because, you know, I know people who serve on the Nobel Committee. They do the best they can. And they are trying to be fair as much as they can. In fact, if anything, on the whole, I would say they are very conservative and they have been conservative historically. You know, after 50 years, they can release the discussions that were on the committee 50 years ago. And I've had the opportunity to look at some of those on the day they were released. And it's incredible how people who we now think of as household names were considered to be rather radical or not having quite established what they were doing, you know, 50 years ago when they were considered. But by the same token, there is, of course, the Nobel Committee does not operate in a social vacuum. And there has and is, I suppose, pressure on them to, you know, promote certain results. So as it happens, I do in fact know the background to how this prize was awarded, but I'm afraid I can't really discuss it because the Nobel Committee does maintain total confidentiality about their process for 50 years. So I'm afraid you'll have to wait till 2061 to really know what went on about that award. But for the moment, all I would say is that it was an important award because it was the first time that—actually not quite the first time, there had been an award earlier for pulsars, for example, radio astronomy—however, it was one of the first awards for cosmology and it was important in establishing cosmology as a physical science and raising its prestige in the community. So I think on the whole, it was a good award. And the fact that some of us are questioning the result today, I think that's just how science progresses. It's not always obvious at any given point in time whether something is beyond doubt. So that's a long way for me to try to evade your question because I don't

want to offend people and, you know, I do have respect for the awardees, so I don't want to diss them. You don't want to diss them. Now I didn't realize that this was as spicy a question. I knew it was a bit controversial, but I didn't think it would have this much background to it. What can you tell me? Well, since you raised it, let me say that I did have one of my papers rejected by a well-known journal. I will not name it, but I challenged the decision because I didn't like the referee report. And the receiving editor basically said, "Well, we can't publish your paper because it questions a Nobel Prize-winning result." And that did annoy me because I don't think any result in science is above being questioned, whether it's for a Nobel Prize or not. Receiving the Nobel Prize is a kind of imprimatur that it is an important result. And they usually are right. But, well, actually, let me relate an anecdote. I don't think Stephen would mind. So Stephen Hawking used to have this, when he was still with us, he used to have this sort of what can only be called soirées in a country house, generously supported by a foundation, and invite people who he had heard were doing something interesting to come and talk about it. And I once was invited to talk about the work we had done on the significance of the acceleration. And having heard it—and he sat through the whole talk, which was, you know, quite terrifying because it would have been very upsetting if he had wheeled his chair out halfway through—but he sat through it and then he composed a question. And we had to wait, you know, a fair bit of time for him to compose the question on his machine. And he finally said, "Has anybody been awarded a Nobel Prize for the wrong reason?" or something like that. Interesting, okay. And actually, I didn't know the answer offhand, but my colleague from Cambridge who had invited me, he did. He said, "Yes, it was Enrico Fermi." And Enrico Fermi apparently was awarded the Nobel Prize for the discovery of transuranic elements, which was not true. They were in fact discovered later. What he was in fact seeing was nuclear fission and he didn't realize it. So of course nobody is ever going to question that Enrico Fermi fully deserved a Nobel Prize for any one of the number of things that he did. But historically, that is a case where a Nobel Prize was awarded for the wrong stated reason. So the one in 2011 was awarded for the discovery of cosmic acceleration, not dark energy. Strictly speaking, they awarded it for the discovery of cosmic acceleration, and that actually I think was not right. You know, there is no cosmic acceleration. We do see cosmic acceleration, but it is only in one direction in the sky; it is not all over the sky. And it has therefore nothing to do with a cosmological constant. It is a local effect, which we think is due to the fact that we are embedded in a deep bulk flow, and the energy density of that flow affects the interpretation of our measurements such that we think we are accelerating when actually it's really a kind of a perception. It is an illusion, if you like. The overall universe might be decelerating. Interesting. Okay, so let's take this step by step. I'd like to build up to various results and even go further into cosmic voids and baryon acoustic oscillations and so forth. So in 2016, your analysis found evidence that when taken on its own, something that was thought to be a five sigma result—which is extremely high, the higher the sigma, the better the result—was actually a three sigma result or three standard deviations. And it wasn't a discovery of; it was more like evidence for. So can you please explain about that? Yes. Well, what we discovered was that for a start, the initial discovery, the two Nobel Prize-winning teams, they essentially had of order 50 supernovae each, right? So a total of 100 supernovae. There was in fact

an overlap in the two samples. And the other thing is also worth mentioning: the two teams are not quite independent. Because this is something that people set store by. We like to see five sigma results from different independent experiments. So for example, the discovery of the Higgs was where the ATLAS and CMS collaborations, they were completely independent of each other and both had five sigma. And that result, as you know, has stuck and it has grown with increasing amounts of data. So, the basic test is, if you have a threshold, you know, if you cross the five sigma threshold, then you can claim a discovery. But actually, as you alluded to earlier, that is not a guarantee that what you are seeing is in fact the case. Sometimes even five sigma results go away. I know of a seven sigma result measured in the laboratory that went away. This was a discovery in 1991 of a neutrino of mass 17 keV. It actually happened at Oxford. I know the story pretty well. And it was confirmed by an experiment at Berkeley and then another one, but then other experiments subsequently didn't find it. And it turned out to be a conspiracy of three separate systematic effects in the experiment that had been done at Oxford. So, you know, the thing is that we don't talk too much about when experiments go wrong, when we get failures; we usually talk only about successes and discoveries. And I think that is actually a—well, this way, not just the public, but even the practicing scientific community, young people, they don't really get the kind of education that they need to understand the nature of scientific discovery. That for every discovery that is made, there are a lot of false alarms, there are a lot of wrong alleys that are explored. And by definition, if you're seeing something, five sigma is meant to be the kind of accidental chance of seeing something at five sigma is three and a half in a million. Yes, the odds are, you know... Yeah, which is pretty small. But the joke in the physics community—this was said by a famous particle physicist—is that half of all three sigma results are wrong. Now, three sigma is 99.7%, right? Yes, interesting. That you think is pretty unlikely. How can 50% of some result which is meant to be 99.7% significant be wrong? Well, that's because of the systematics that you alluded to earlier, right? Because this three sigma refers only to the idealized case when you do not consider systematics; you consider only statistical fluctuations and you believe that those fluctuations are governed by Gaussian statistics. So you have the bell curve and you're looking at the area under the curve, and one, two, and three sigma and going to five is just the area under the curve. But so it's a convenient kind of language we talk about, but we have found empirically, at least in particle physics, that five sigma results generally tend to lead on to results that are established beyond doubt subsequently, especially when obtained by two independent experiments, right? This is the rule of thumb. It is not a theorem. In principle, as I said, there can be cases where even seven sigma results have been wrong. But usually two five sigma results independently obtained are generally considered to be the threshold for discovery. You spend your days wrestling with the vicissitudes of consciousness, free will, and whether reality is a simulation. But some problems have elegantly simple solutions. BlueChew Gold is the newest innovation from the number one chewable ED brand. And gentlemen, this is a four-tablet which is setting the gold standard for performance. Two ingredients for blood flow, plus apomorphine and oxytocin to amplify arousal and connection in both brain and body. It dissolves under your tongue and can work in as little as 15 minutes. Look, my wife didn't marry me for my humility. She married someone who optimizes

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All right, so coming back to these supernovae, there had been two teams and they had seen about of order 50 supernovae each. What we discovered subsequently that was interesting is that they had mainly been looking at just one part of the sky. And that is the part of the sky towards which we see the cosmic microwave background is slightly hotter by about one part in a thousand than the opposite direction. And that we believe is because we are moving towards that direction. And the reason we are moving is because the universe today is inhomogeneous. If the universe was homogeneous, then every direction would be the same. There would be nowhere to move to. But because the universe is inhomogeneous, we have local attractions and even repelling motions, which are not Hubble flow. The Hubble flow is radial, smooth. Every galaxy is supposed to be moving away from all other galaxies. But in practice, if you look very locally, that is not true. So if you look at our nearest galaxy, which is Andromeda, it's actually falling towards us. It has a, not a redshift, but a blueshift, right? And actually, we are falling towards Andromeda. We are falling towards each other because we are bound in an orbit. And in fact, we'll merge in about 5 billion years, right? That's not the Hubble flow. However, the belief is that if we average on large enough scales, then we start seeing the Hubble flow. So if you imagine drawing contours of velocity around us, they'll be very ragged in our local neighborhood. But as you go further and further out, they'll become smoother and smoother, nice circles. And the scale at which this is supposed to set in is believed to be of order 100, 200 megaparsecs. A parsec is 3.3 light years and mega is million. So you'll have to go out to that kind of a scale. And I'll tell you later how that number is obtained. But that's supposed to be the expectation in the standard cosmological model, which is also a theory of structure formation. However, we find that—so therefore this peculiar motion that we have, which is causing us to see half the sky as hotter than the other half—that is not unexpected. In fact, it was predicted. So Dennis Sciama, the cosmologist that I first worked with, who is known for his very famous students like Hawking, and he also turned Roger Penrose on to cosmology and had many other brilliant students—so Dennis actually wrote a paper in '67, which said of the newly discovered cosmic microwave background, “Well, I think you will find that it has a dipole anisotropy; half the sky will be hotter than the other side because we have a peculiar motion. We are not in the frame of reference in which the expansion is isotropic, in which all the galaxies are going away from each other.” That paper, by the way, I looked it up recently. It had about 20 citations since 1967. Somehow the most profound papers often are overlooked. This is another kind of aspect of real life. Interesting. But of course, that dipole anisotropy was found subsequently. And it was, since it had been predicted as due to our kinematic motion, and what we saw matched it. So, Jim Peebles and David Wilkinson at Princeton, they actually calculated what sort of a temperature variation we should see across the sky if it is entirely due to our motion. So you can calculate it because that is just special relativity. You know, today we

would set it as a first-year problem, how to calculate this thing. Anyway, they did all that. So that was established. Now, it is interesting that the supernovae that were looked at by the Supernova Cosmology Project and the High-Z Supernova Search Team, they were also looking mainly at supernovae in the direction of the hotspot, the direction towards which you are moving. And that's interesting because that means that they are looking at not with the whole sky, but only at one part of the sky. And according to this idea that I mentioned earlier, that we are in a deep bulk flow, which is actually moving in roughly the same direction, we should be seeing acceleration only along the direction we are moving. So if the two teams had actually had the opportunity to look over the whole sky, they would have found that the overall expansion is actually—in fact, it's a dipole. We are seeing acceleration in one direction and deceleration in the opposite direction. If you average over the whole sky, you will find something pretty close to expansion at a constant rate, which used to be called the Milne universe, the Milne model, because Milne had a kinematic theory in which the scale factor just increases proportional to time. And so, what we found in this paper in 2016, using principled statistical methods, is that the expansion is actually pretty close to expanding at a constant rate, neither accelerating nor decelerating, if you average over the sky. But subsequently, when you looked at it as a function of what direction you are looking in the sky, then it became evident that it was actually a dipole pattern. The acceleration is in this direction; opposite direction, there's deceleration. But that you could not see until we had supernovae covering the whole sky. And that was only—that catalog was only released in 2014. It was called the Joint Light Curve Analysis catalog of 740 supernovae. And today the latest catalogs, there's one called Pantheon Plus Union 3, they have of order twice that number, you know, 1500, 1600 supernovae. So now that dipole pattern is pretty clear. And to go back to the question you asked then, what we found was that the acceleration which had been claimed could only be—the data was consistent with acceleration only at three standard deviations. In other words, it wasn't the kind of discovery level of evidence. Okay, so this is almost exactly 10 years ago. We're now in 2026, and this was in 2016. I recall Rubin and Heiden, if I'm pronouncing their names correctly... Yeah. They said that what you did—and I think it was the same year, 2016 as well—so that you used a profile likelihood and what you should have done, because you had many nuanced parameters like light curve shape and color and host galaxy masses, was Bayesian marginalization, which is the more correct approach. And they redid your analysis with what they considered to be the proper analysis, the proper marginalization. And they recovered a larger than 5 sigma or 5 sigma. So do you think that they got something incorrect or what? Actually, the irony is that Rubin uses exactly the same statistics that we did. They use what they call a Bayesian hierarchical model, but the underlying engine for that is exactly the same maximum likelihood method. It's just dressed up in Bayesian language, but it's exactly the same technique. In fact, they recovered a result—so in that paper, they actually first do the analysis the way you do it and they recover the result. Then they say something else which recovers the significance of five sigma again. And that is, they say that the light curves were adjusted. This was done by the Joint Light Curve Analysis people. They are the astronomers. So every astronomer in the world who was doing supernova cosmology was on that JLA paper, including the Nobel Prize-winning people that I mentioned

earlier. And what they said was, “Look, different catalogs have used different techniques and it’s hard to kind of combine the data, so we’re going to take all the data, whether from the Sloan Digital Sky Survey or something called the Supernova Legacy Survey which was four directions in the sky.” The SDSS was just a strip on the sky. Or there was a dozen supernovae from the Hubble telescope, right? There was a so-called GOODS survey. Not very many, but they were important because these are at high redshift. So you can’t see them from the ground. You have to go out into a satellite to be able to get above the atmosphere to see this because they’re in the infrared. They’re so redshifted. And then there are about half the sample, half the catalogue, is actually very local supernovae within a few hundred megaparsecs. They are all over the sky. So, there are four catalogues which they combine together. These four catalogues have different distributions in redshift. They have different sky coverages. But at least they were all analyzed using exactly the same light curve fit, which they called a Spectral Adaptive Lightcurve Template, SALT. It’s called SALT 2. Okay, that was the thing. And that essentially is the technique by which supernovae can be used to do cosmology. So supernovae by themselves are not standard candles. They vary a lot in their intrinsic luminosity. But it was found empirically, in fact by an astronomer called Phillips, that the peak luminosity of the supernova is correlated with the width of the light curve, right? Purely empirically found this. And actually, finding that correlation is a crucial clue to supernova cosmology. And I might remark that this was only made possible because the astronomers, including the Nobel Prize-winning astronomers, got this breakthrough technique of surveying the sky with CCD cameras. So what then happens is that you look at the sky and then you go back and look at it again the next night and next night and occasionally a supernova goes off. But then you have the luxury of going back in time to see what that patch of sky looked like two weeks ago when it was just starting to rise. You see, ordinarily this would never have been caught because you just see something which has just exploded. You catch it at maximum, but you don’t know how it got there. Yes. But thanks to the CCDs, you can. And that means you get the whole light curve. And that means you can measure its width. This was the real kind of breakthrough. And then they discovered that there was a correlation between the peak luminosity and the width. And using this, you could correct for that scatter in the supernova absolute brightnesses and fit them all onto one template. You could stretch the light curve, which is called a stretch correction. And the stretch correction is different in different color bands. So there’s a color correction, right? Because you observe all of the supernovae in three different bands. And so you have a handle on their spectral qualities as well. So the light curve template fitter simply says: you measure a supernova, it’s got some magnitude—in other words, some amount of light coming out of it—you also measure its redshift, if you can. That is the other quantity you can measure. But that magnitude can be corrected for the observed width of its light curve in three different bands. And by doing that, magically, the scatter, which was a factor of 10, can be reduced to less than a factor of two, right? And then they are so-called standardizable candles. They’re not standard candles, right? So this part is usually glossed over. I have to tell you, you know, I’m not a—I didn’t know about any of this. We had to teach ourselves all this stuff because this is pretty hardcore astronomical, you know, a lot of rigorous and very careful work done by a lot of people.

And one needs to learn what they are doing and why it matters. However, we took these corrections which had been tabulated by this JLA team and they had done the work. So we took it as they gave it. What Rubin and Heiden said was actually these corrections are themselves dependent on redshift. These corrections are not universal properties of supernovae, but they depend on the sample and they depend on the redshift of the object, right? Okay, so what though? Sorry, so these stretch and color corrections that had been given to us, which we were using, they said that we had taken them to be constant with redshift because that's actually what the collaboration had told us in the paper, but that actually if you looked at the corrections carefully, you could imagine that they in fact varied with redshift, right? So they introduced 12 more parameters. In our fit we had 10 parameters, and they doubled the number of parameters. Okay. I have said if we allow for that, then in fact the significance of the acceleration again goes back to, you know, something close to five sigma. This was their argument, right? And the reason why we didn't go along with that is because, you know, if you add enough parameters to a problem, you can, as Oppenheimer famously said, "You can fit an elephant." Okay, so technically you're allowed to add parameters if it improves your fit substantially. So there is a criterion. Roughly speaking, if you add one parameter, your chi-square per degree of freedom should decrease by two, you know, twice the number of parameters that you are adding, something like that. More formally, there is something called a Bayesian Information Criterion. So there are ways to check on this. And we didn't like the fact that they were adding so many parameters. But more to the point—and this is important—if they're saying that the light curve properties can change with redshift, then why can't the absolute magnitude of the supernova change with redshift? Right? I mean, if you are saying that the intrinsic properties of a supernova are redshift dependent, that would completely undermine their use for cosmology if the absolute amount of light they are putting out also had to change with redshift, right? So you're kind of opening a Pandora's box there. You really shouldn't do that, right? So then what is their response to your counter paper? Well, we wrote a counter paper, but we didn't feel it was—no, we don't like publishing umpteen papers. It is a note on the arXiv and we showed quantitatively that what they were saying is not actually valid from an information-theoretic point of view. So we didn't take it seriously because if they thought about it themselves, they would have realized that actually they can't just say that we are going to let the stretch and color corrections be redshift dependent. Because that raises the possibility that the absolute magnitude is also redshift dependent. And in fact, 20 years later, that crow has come home to roost because a group of Korean astronomers have found that there is actually a correlation between the absolute magnitude of the supernova and the age of the progenitor, the star that blew up, right? And in fact, what they find is that as you look back in redshift, you're looking at typically younger objects, right? And they would be intrinsically fainter, which is just the effect that you are ascribing to cosmic acceleration. Okay. So in other words, it looks like other astronomers following that train of thought have actually now been looking into it and claiming that the absolute magnitude of supernovae, the thing that is supposed to make them standard candles, is in fact not a standard candle. It is correlated with the property of the host object that blew up, the star. And if the star was younger, then the supernova would be intrinsically fainter. And

in fact, when we take that Korean group's suggestion and implement it to our analysis, it is evident that actually on large, very large scales, there is deceleration. There is no acceleration. All the acceleration is happening locally and it is dipolar. So it can't be to do with the cosmological constant. Now, I'm sorry, this is getting rather technical, but I'm afraid it is true that most people don't actually follow these arcane details of supernova cosmology, right? They just are told the simple story that you are looking at exploding stars and they have put out exactly the same amount of energy wherever they are in the universe. So by just observing them, we can tell how far away they are. And then if you can also measure the redshift, the redshift versus the brightness determines the cosmology and that requires that the universe is accelerating, right? That is the story we're told. But the statement, the real statement, is the following. When we look at distant supernovae, we see them to be about 0.3 magnitudes—which is roughly 30%, it's a little more—fainter than they would be if the universe was expanding at a constant rate. That's the statement. So then you have to ask yourself, if you are a physicist, this 0.3 magnitudes, which is the difference between the accelerating case and the non-accelerating case, right? How does that compare with the kind of scatter, the fluctuations, the kind of corrections that you are applying? If you're applying a correction that is 0.15 magnitudes, then you can't really claim that a 0.3 magnitude is a big deal, right? I mean, all your corrections, all your adjustments should be much smaller than the effect that you are claiming is the evidence for new physics, something as startling as cosmic acceleration, right? And I'm afraid it is the case that now it is becoming quite evident that the evidence for acceleration is actually very weak. In fact, as more and more data comes in, it is becoming more and more evident. This is in total contrast to the case where you start by seeing a little bump in, you know, some cross-section in some laboratory reaction. It is not significant, but you take more data and the bump grows, and you take more data and the bump grows, and then one day the bump is so strong that you can't deny it; it is there, right? Yes. Now that is the traditional way particle physics is done, and that's what's called frequentist statistics. You use things like p-values, you know, you ask what is the random chance—you have some null hypothesis—what's the random chance of the data that you are getting really. The point being that you can only disprove a null hypothesis, okay? It's very hard to prove anything, right? You can prove things wrong. This is much simpler to do and actually, philosophically, that's what we do. You know, we disprove things wrong. We take a null hypothesis. So, the null hypothesis in the other case would be that the universe is expanding at a constant rate. If you can prove that wrong, then you have a result, right? The Bayesian part comes in in cosmology because you don't have the luxury there of repeating the same experiment over and over again. There's just one experiment. It happened. Okay, you got one sample of data. You can't necessarily repeat the same experiment, identical experiment over and over again. So, the whole premise of these sigmas, the Gaussian distributions, etc., that is really a bit dodgy. So, in cosmology, what you do is this Bayesian analysis where you—Bayesian statistics and frequentist statistics should give the same answer if you are asking the right questions. Okay, there's no—statistics can't change the physics. But in some situations, Bayesian is much better suited than frequentist, right? For laboratory work, frequentist is perfect. You know, I'm on the Particle Data Group. We maintain this review of particle properties, which is like

the Bible of particle physics. Everything in there is quoted according to frequentist statistics. Because if you had to use Bayesian statistics, you could do Bayesian statistics, but then you have to state, along with the result, you have to state all your prior assumptions. Right, right. Okay. Now, Bayesian statistics is, in an ideological sense, better. I agree with that because it... Why is that? Because it makes explicit that everything that we say is subject to some prior assumptions we have made. Okay. We have some priors in our head, you know, biases, if you like, right? Yeah. But I don't see why that's superior to a frequentist approach. Because the frequentist statistics works fine under ideal conditions, but in non-ideal conditions, it is sometimes helpful if you state exactly what you assumed as the prior distribution of your variables in order to arrive at the answer that you gave. In the case of frequentist statistics, you have the luxury of getting more data. So to give you an explicit example: this Higgs boson. Okay, think of the Higgs boson. So people were looking for it. We didn't know where it was. It was somewhere between 100 GeV and 900 GeV. That's all we knew, right? So you look for it. And then they said, "Well, there is a little bump at 125 GeV." It was initially just three, three and a half sigma, right? But it was seen by both ATLAS and CMS. And at this point, a lot of cosmologists started shouting, "Oh, you know, these guys, they don't know about the look-elsewhere effect." The point is that once somebody says there is a bump at 125 GeV, if I'm doing another experiment, I would look there. But that's a kind of a bias. I should be looking everywhere, not just there. But the point is, that is what you should do. If you're doing cosmology, you should do that. You should not just focus on what somebody else has said. You should be unprejudiced. Freeman Dyson actually talked about how he's happy that the Higgs was discovered, but he's unhappy about how it was discovered because of this exact reason: that you have to throw away plenty of the data. And so you know what you're looking for and so you look there. That's right. But the proof of the pudding is the eating. You see, they took 10 times more data. The peak was not only there at 125—it didn't go away—but it got higher. And it got to 5 sigma in both experiments. And at that point, it was pretty clear that it was the Higgs. But data has continued to come in. And, you know, lots of properties of the Higgs have been measured. There is absolutely no doubt in our minds today that there's something there. However, this may not have been so. Subsequent to the Higgs discovery, there was a false alarm that there was a bump at 740 GeV, right? Hundreds of papers—actually 740 papers—were written trying to explain this bump, right? And it was, if I'm not mistaken, it got to over four sigma in both ATLAS and CMS. That was what was misleading most of the theorists because, you know, the rumors were flying left and right and everyone heard that both experiments had seen it. So it came to over four sigma in both experiments and then it went away because such things can happen. You can have fluctuations. I mean, nature is indifferent. Nature is not either good to us or bad to us. Nature doesn't care. Nature is tossing dice. And sometimes you can have a situation where, you know, the wheel at Monte Carlo can come up red 10 times in a row. It's not a biased wheel. It is a fair wheel. At least we think so. But it can happen. The odds of that is just half to the power 10. It's not that small. You know, 2 to the 10 is 1 over 1024. It's 1 in a thousand. It's hardly anything, right? The point I'm making is that, in principle, you cannot ever tell the difference between a biased coin and a very rare event where the coin comes up, say, heads. You can never tell the difference. That's

just a matter of your attitude. So if you're a mathematician, you would say every time you toss a coin, the odds are one in two, heads or tails, right? This video is sponsored by BetterHelp. If you've spent any time with the questions that we've explored on this channel, like consciousness, free will, the hard problem, the nature of existence, you know how demanding sustained intellectual inquiry can be. The mind doing the investigating is the same mind that needs care. BetterHelp connects you with a licensed therapist entirely online, making professional mental health support accessible regardless of your schedule or location. You fill out a questionnaire about your needs and preferences, and you're matched with a therapist, typically within 48 hours. If the fit isn't right, then you switch and it's extremely straightforward. There's no additional cost, no awkward talk about ending that relationship. There's no commute, there's no waiting room, there's just a flexible format that works for your life: video calls, phone sessions, or text messages. Whether you're wrestling with the large existential questions that are on this channel or practical difficulties of daily life or grief, loss, sorrow, depression, having a trained professional to talk to makes a huge difference. If you're considering therapy, visit [BetterHelp.com/CurtJaimungal](https://www.betterhelp.com/CurtJaimungal) and get 10% off your first month. That's [BetterHelp.com/C-U-R-T-J-A-I-M-U-N-G-A-L](https://www.betterhelp.com/C-U-R-T-J-A-I-M-U-N-G-A-L). Let me make this clear. So my understanding is that you were not claiming that dark energy doesn't exist. It was just that the five sigma result should be a three sigma result. But now it sounds to me as if you're claiming that you believe dark energy doesn't exist. That's correct. Okay. So I want to know what changed, how it changed. But also I want to know about this variable dark energy results from DESI DR1 and DR2 about its weakening. So does that comport with your worldview? Does that take a sledgehammer to it? So tell me how you think about all of it. Okay, so to answer your first question: that's right. We started doing this analysis when the catalog was first made public. The supernova data was made public in 2014. It actually took us a year and a half to get our paper published because there was a lot of pushback. I mentioned earlier a famous journal that has turned us down because it was a Nobel Prize-winning discovery. But anyway, it did get published in Scientific Reports in Nature and caused a bit of a stir because, you know, we could show using rigorous statistics that it was not a significant result. But that was just the start of it. So as you said, then there was this pushback by Rubin and Heiden saying, "Well, we should allow for the supernovae to evolve with redshift." But subsequently, when we looked at the distribution of this acceleration on the sky—so normally we analyze all data in cosmology assuming that the universe is isotropic, okay? And that the expansion rate, you can do a, if you like, a Taylor expansion; you can say it's got a velocity, then it's got acceleration, then it's got a third derivative, that's called the jerk, and so on. You can do an expansion like that. We allowed for this acceleration to have one more degree of freedom. We said, let it have an angular dependence. Let it be a monopole plus a dipole, right? We didn't add more terms because you want to minimize the number of parameters. We have only 740 supernovae. We don't want to increase the number of parameters too much. But then we let our maximum likelihood estimator loose on this data set. And it told us that actually the data overwhelmingly thinks that the acceleration is a dipole. It is only in that direction. And it's negative in the other direction. And basically, it's a dipolar pattern on the sky. So when we published this result—this was in 2019—that caused

even more of a sort of hullabaloo. And I recall Adam Rees, who is one of the Nobel laureates, he told Physics World, “Well, these guys, they’re using, you know, old data,” although he was actually one of the authors of this 2014 data set, “and we have many more supernovae now and also there is other evidence that dark energy exists.” These were his two main arguments. So with regard to the first statement, indeed we are using the old data set because we thought it was a good data set. But now there is a new data set called Pantheon Plus of which Adam Rees is a leading author. So we have taken that data set and redone our analysis from 2019 on it. And that paper was published last year in 2025, right, after rigorous refereeing. And we find that the dipolar signal is stronger than ever. It’s even stronger in the 1700 supernovae or so in the Pantheon Plus data set. Why is this? And—oh, that’s the first part of the statement. And now, if we allow for the supernovae to also have a luminosity that depends on their redshift, depends on the age of their progenitor star, which the Korean astronomers are saying, then in fact, as we go out, it’s a dipole, the dipole dies out, but there is still a monopole component left. There is an isotropic component. That isotropic component also disappears if we allow the supernovae to have a luminosity that depends on the age of the progenitor. And the universe then looks like it is actually decelerating. But locally, it looks like acceleration because it is aligned with our local bulk flow. And it could be just a local effect because of that. So now I’m of the opinion that, in fact, there is no cosmic acceleration at all. And that certainly is not evidence for dark energy. Because something that I should stress is that a cosmological constant, vacuum energy, has to be isotropic. Because otherwise you violate Lorentz invariance. You know, different observers will not see the same vacuum. And that is sacrosanct. The problem is—and I should make a strong statement about this—I really get very upset when I see people, astronomers, especially in supernova cosmology, fit things to supernova data without putting in any physical priors on it. In other words, prohibiting things which are unphysical. So they consider the possibility of equations of state which are so extreme that you would violate unitarity. You know, you would violate things that are absolutely sacrosanct. But it is allowed in their fit, and the data—one discrepant data point can pick out something like an equation of state which violates the strong energy condition, for example, right? But there is an unfortunate gap between relativists on the one hand, astronomers on the other hand, and particle physicists, right? There’s a gap? Meaning what? Well, the culture gap, both a gap in the background and the way that one looks at phenomena and the ability to appreciate that a cosmological constant, for example, to an astronomer is just a number. It’s like, you know, Ω_Λ is 0.7, right? And they think it’s a simple model because they consider it to be one parameter in a six-parameter or seven-parameter model. They don’t appreciate that for Λ to have that value it does—which is something of order the present-day Hubble parameter squared, right?—to interpret it as vacuum energy, you would have to adjust operators in the standard model to 60 places of decimals, right? You have to cancel terms against each other until everything is gone except one number, one in the 60th decimal place. Because the natural scale of the standard model is a TeV and the energy scale of this dark energy is an eV. So there is, you know, a lot of gap. There’s a huge hierarchy between those two scales. And it simply makes no sense to call it something to do with the vacuum because it’s got nothing to do with the standard model. There are no energy

scales in the standard model of order the vacuum energy scale. And in particular, it cannot have anything to do with anything because the scale is set by the present-day Hubble parameter, which is what enters into every measurement, right? But the Hubble parameter is neither fundamental nor a constant. Why should it determine the cosmological constant? I mean, I really don't get why more people are not, you know, struck by this. That's what I wrote in my essay. I said, "At this point, alarm bells should be ringing." The Heart of Darkness essay? Yes, that's right. That is the main point of it. Just for people who may be tuning in and wondering what are you referring to when you talk about this gap and where do you, Professor, lie on this: you from 2011 to 2019, I believe, were the head of Oxford's particle theory group. That's right, yeah. So you have a perspective on cosmological issues that someone who is a pure relativist or a pure astronomer wouldn't have? Well, I kind of see the points of view of all these three groups. I kind of, you know, I can see how each one looks at them. But I find it, as I said, somewhat frustrating that people who fit the data don't appreciate that sometimes they are playing havoc with fundamental rules of nature. They're violating sacrosanct things like unitarity, which you cannot violate, you know, however radical you are. Well, if you were to violate that and you were able to show it, then that itself would be worthy of a Nobel Prize. That would be astounding. But, well, I would very much doubt because in that reality, our universe would not exist. So no Nobel Prizes would exist either. But that's another matter. What I'm trying to say is that, just to put it very simply, if I am analyzing data in the framework of a model which forces the only unknown quantity, namely λ —which is a quantity that is allowed to be added in the standard cosmological model, simply reflecting the fact that we have assumed the model to be strictly isotropic and homogeneous—so it has only got three parameters. There is a matter content, there is a curvature of spatial surfaces, and then there is this quantity λ , which simply reflects the underlying symmetry of general relativity that you have local coordinate invariance, right? And then in the, you know, when field theory was discovered—I mean, this was done at a time, and Einstein sort of got all this, when we still thought of matter as particles, non-relativistic particles. Of course, later we realized that if you heat up particles enough, they become radiation, but that's still matter. It has got pressure, but the pressure is positive. However, in the 1930s, when field theory was discovered, Wolfgang Pauli, I think he was the first guy to realize that the ground state, the zero-point fluctuations of all these quantum fields, also act like a cosmological constant. And the magnitude of these fluctuations is huge. I mean, he looked at the fluctuations that would come just from electromagnetism, quantum version, right? And he wrote that if this coupled to gravity, then the universe would not even reach to the moon. It would be prevented from becoming as large as it is because of the cosmological constant of the vacuum. In fact, he actually made a calculational mistake. The answer is that it would not have been any bigger than this, but bigger than a few millimeters. And he, in fact, wrote—this is in the Handbook of Physics, 1933—"As is obvious from experience, the zero-point energy does not couple to gravity." Okay. Now, this is the so-called cosmological constant problem, which, you know, Steven Weinberg memorably called—he said it was "the bone in our throat." The catastrophe. Yeah, it just defied explanation because according to general relativity, which is a classical theory after all, all forms of energy density must couple to gravity, right? But as

Pauli said, “As is obvious from experience, zero-point energy does not couple to gravity,” because if it did, then the universe could not ever have expanded to be as large and as old as it is today. Professor, I don’t see how this helps your case. So help me understand. Firstly, is there a reason to believe that dark energy—that we should get to a calculation of what its value should be from vacuum fluctuations? Is there a reason to tie those two together? Not necessarily. I know naively we do a calculation, so we’re 10 to the 123 or something like that, or orders of magnitude off. But is there a reason to tie the vacuum fluctuations to dark energy? Well, strictly speaking, in field theory, you can never actually calculate the absolute value of the vacuum energy. You can only calculate differences in vacuum states, not absolute values. Technically speaking, it’s a so-called super-renormalizable term in the Lagrangian. So you can’t actually calculate. But what you can say is that without fine-tuning, its value should be set by the energy scale up to which the theory is valid. And we believe that our standard model is valid to a few hundred GeV energies at least. Statement one: we can’t actually calculate the vacuum energy, but it should be something whose scale is set by a few hundred GeV. Call it a GeV. Statement two: how does this vacuum energy affect the expansion of the universe? That’s the second question you can ask. The naive answer is it would have stopped the universe from ever expanding beyond the size of a nucleus. As the universe cooled down and it got to a temperature of a TeV, so the thermal energy has gone, all that is then left is the vacuum energy that would have either caused the universe to re-collapse or to go into eternal inflation, one or the other, depending on the sign. And we can’t even calculate the sign in field theory. The very fact that that did not happen, that we are here discussing it, seems to suggest that vacuum energy does not couple to gravity, as Pauli correctly noted, I think, in 1933. However, we do not understand why that should be so, because according to Einstein, everything couples to gravity. So that, in a nutshell, is the essence of the cosmological constant problem: that our best theory of gravity, which is a classical general relativity, does not, when coupled with the fact that you have the quantum field theory of the other fundamental interactions—at their interface, at their very uneasy interface, is this cosmological constant problem. If vacuum energy coupled to gravity, we should not be here. But we do not know any reason why it should not couple to gravity in the formulations that we currently have of these theories. And my biggest disappointment with all attempts to construct theories of quantum gravity is that none of them—you know, loop quantum gravity, string theory, whatever you name—none of them have been able to address the cosmological constant problem. They usually evade it.

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only have daily articles, but they also have long-form podcasts with their editors and writers. This is also available online. Whether it's scientific innovation or shifting global politics, The Economist provides comprehensive coverage beyond headlines. As a TOE listener, you get a special discount. Furthermore, until January 18th, just this week, it's 50% off. Head over to economist.com/TOE to subscribe. That's economist.com/TOE for your discount. Doesn't string theory address it with supersymmetry? Well, the point is that the real world is not supersymmetric in any case. Even if the world were supersymmetric, it would be violated at an energy scale of a TeV and you have the same cosmological constant problem as before, right? So that's the simple answer to that question. The real answer is a bit more involved, but that should do. So there is no way that they address the cosmological constant problem. And therefore we are missing something really big, right? Every attempt to sort of solve the cosmological constant problem has failed. There are actually now review papers classifying all the different ways people have attempted to solve it, many clever and intricate ways, and they have all failed, which means we are missing something really big. And I like to tell young students, they should not worry that all the big problems have been solved and there is nothing for them to do because we have missed solving the biggest one of them all, right? Now into this situation—so what happened? Particle physicists realized they couldn't solve the cosmological constant problem, but then it doesn't really affect them. It affects nothing that is done at the LHC, okay? This is something that only affects stuff on the larger scales, on the scales of the universe. So as I put it in my essay, it was an accident waiting to happen that when you start making observations of distant objects and you allow for a cosmological constant in your equations, you are going to find it to be non-zero because you are fitting the data in a very constrained framework where the only unknown quantity in your equation is the cosmological constant. You have matter, you have curvature, and you have the cosmological constant. You can measure the matter independently. You can measure the curvature independently. And you determine the cosmological constant from what's called a sum rule, which is that the fractional energy densities in matter, curvature, and cosmological constant add up to one. This is a restatement of the Friedmann-Lemaître equation, which is the workhorse of standard cosmology. And that equation is derived assuming exact isotropy and homogeneity, the so-called Friedmann-Lemaître-Robertson-Walker metric. And that leads me to answer your second question. I'm getting there. You asked about evolving dark energy at DESI. DESI has got some brilliant data. It's got millions and millions, you know, 10, 12 million spectroscopic redshifts measured with this beautiful instrument they had on this telescope at Kitt Peak. And it covers almost, I don't know, 35% of the sky by now, right? I think they get to about 40% of the sky eventually. They are still analyzing this data, this beautiful data set. The first equation in their paper is: "We assume the Friedmann-Lemaître-Robertson-Walker metric." They assume exact isotropy and homogeneity to analyze the data. Which is the very thing you're questioning. Yeah, which I'll come to in a second why I'm so doubtful about it. But they allow an extra degree of freedom now. They say it's dark energy instead of just being something which has got a purely diagonal energy-momentum tensor, it has got a second term which is proportional to redshift, right? This parametrization is actually without any physical basis whatsoever. It is simply for calculational convenience. It's

self-serving. It's so that you can integrate observational quantities like the luminosity distance or the angular diameter distance in a neat way because, you know, it's in the integral. So if you make it proportional to Z , then you can integrate over Z . It's just that. And then they claim that it differs from zero. They have got evolving dark energy. My question, which I've asked DESI people several times, is: "Have you checked that your signal that you're actually measuring, like the BAO signal, is it the same in every direction in the sky?" I mean, you've got like 12 million data points. Are they indicating a peak at exactly the same position whichever direction you look over the bit of sky that you have looked at? And I have not had any answer to the question. It is certainly not stated in any of the papers. In other words, they are analyzing the data according to an assumption of a metric that was actually made over a hundred years ago. It was in the 1922 paper by Alexander Friedmann. He assumed the same metric because at that time we had no data. We didn't even know we lived in a galaxy, right? Lemaître assumed the same thing, and Friedmann and Robertson and Walker then gave the mathematical basis for it, but it was exactly this metric. So what is your metric of choice then? Is it LTB? Well, I don't know. As I said earlier, we can only prove things wrong. It is much harder to find what is right. So what we have done more recently—and this is something I'm very excited about because this is free of the complications that we discussed earlier about the supernova data and all this technical stuff about stretch corrections, light corrections, progenitor age dependence, and so on, messy stuff. People can argue back and forth and you never get to a consensus. I'm now going to tell you as briefly as I can about something. It's so simple that it was actually proposed 40 years ago. It was proposed in 1983 and published in '84 by George Ellis, who is a well-known relativist from South Africa, and John Baldwin, who was the head of the Mullard Radio Astronomy Observatory at Cambridge, at Cavendish Lab, right? And he was undertaking the Cambridge Survey of Radio Sources. And what these guys realized—I was quite intrigued to find that they had actually met at an ex-monastery on the island of Crete where they hold conferences, and this is called the Orthodox Academy of Crete. And they are clearly—I can just imagine them, I've been to that place—chatting, you know, about this thing that Baldwin was about to start counting radio sources over the sky and Ellis says to him, "Look, if the microwave background has this dipole anisotropy due to our motion, then shouldn't any distribution of objects from the sky at large distances have the same anisotropy?" Because that is arising due to, well, physics. It's arising due to the phenomenon of aberration. So aberration is the phenomenon that when you are moving, a star will appear to be displaced in the direction of our motion. And this was actually—it's a relativistic effect, but it was actually found by an astronomer called Bradley. In fact, he was the professor of astronomy at Oxford. Over 200 years before Einstein, he actually measured—said that the angle at which you see the star is related to the true angle where the star is at by a formula, which is just a relativistic formula, taking into account that light has finite speed. So Bradley actually worked out to within one percent the speed of light by observing that this aberration was, you know, so many minutes of arc in the sky. Right. So aberration causes a uniform and isotropic distribution of the sky to be sort of focused in the direction you're moving. And you have to also allow for the fact that when you're moving, you see an object at a different frequency than when you are not moving. So if it has a spectrum,

you have to allow for the fact that there is going to be a different number of objects in that window, etc. So when you do all this, you find that you get a dipole pattern on the sky, which is the same as the CMB dipole. It is V over C . It's a relativistic effect. And V over C is about 10 to the minus 3 because we are moving at a few hundred kilometers per second. In fact, 369 kilometers per second to be precise. And if you do all the maths, then for the case of objects from the sky, like radio sources or quasars, it's actually enhanced a bit. It's about 5 times 10 to the minus 3 , right? So what is it? 0.5%. Okay. It's a tiny, tiny effect. But, you know, to measure something like that, you need at least a million objects on the sky. Now, in 1984, when they published their paper, there weren't a million objects known on the sky. When they published—Ellis? Ellis and Baldwin. So they published their paper in Monthly Notices of the Royal Astronomical Society in 1984. But the first sky map of radio sources that was approaching that kind of a number came out from the Very Large Array in Socorro, New Mexico, around the millennium. It was published in, I think, 1998 or '99, right? And this catalog is called NVSS. This had more than a million radio sources. But actually you've got to throw a lot of them away because you need uniform seeing over the sky. You are looking for the number of objects above some threshold as you are going in one direction, which will be within your band. So as you are moving, objects which would have been too faint to be seen in your survey would be boosted above the threshold, and opposite in the reverse direction. So you'll see a hot spot and a cold spot. But to do this right, you really need a very controlled survey. You need to have the same sensitivity all over the sky. And, you know, when you do counting from an antenna on the ground, as you get closer to the horizon, your sensitivity changes. There are ionospheric reflections, organs of experimental uncertainties enter into the thing. So you've got to cut out quite a bit of your data, right? So when you do all that data cutting, etc., you aren't really left with a sufficient number of sources. But people did this and they said that, "We are seeing a dipole. It is in the same direction as we expect, but its amplitude seems to be off. It's a lot higher than it should be," right? So we did this exercise ourselves in 2017. In fact, one of the criticisms was that NVSS sees only the northern sky because from one position on the ground, you can't see the whole sky. You can only see about 40% of the sky, right? But there is another telescope in Molonglo in Australia which sees the other half. So we combined the two surveys and we made a full sky map. And we saw this dipole and it was indeed higher than expected. But the same sigmas strike again. We did Monte Carlo simulations. We ask, "How many times would we see something like this by chance, by pure chance?" right? And the answer to that question is 2.7 sigma. This effect is not significant at all. It is even less than 3 sigma, right? So, you know, we have to be consistent. We are accusing other people of not having a significant result; the same applies to our result. So we can't claim that it is significant. So it's a curiosity, but we realize we need better data. And then great fortune struck. We happened to meet up with an astronomer. His name is Nathan Secrest. He's at the Naval Observatory in Washington. And he actually serves on the committee that maintains the reference frame, which is used for satellite navigation. And these guys really need to get everything absolutely right. So they use quasars, you know, which are the most powerful sources in the sky. You can see them out to a redshift of a few. They're point sources. They're initially called quasi-stellar sources. That's why they're called quasars.

And he had a catalog with, you know, a million and a half of these guys, right, measured all over the sky by a satellite called WISE, which was the infrared satellite, which operated for years and years. We were fortunate to meet up with this astronomer who, in fact, had a catalogue of quasars mapped in the infrared. So quasars are basically supermassive black holes at the centres of galaxies that are gobbling up matter and spewing it out in jets. But the ones that we are looking at are in the infrared. So, in fact, they are really the accretion disks around the central black holes, right? Similar objects shoot out these beams and they become radio sources. So, you know, the radio sources and the quasars that you are looking at do have a connection, but the catalog that we are using has got no radio sources in it at all. It's a completely independent catalog, right? The way that they are cataloged is completely different. So a different class of objects is picked out in the survey. The quasars that we are looking at have been mapped by this satellite all over the sky. But again, we have to make various quality cuts because we need a map of these quasars on the sky which gives us how many quasars there are per unit pixel on the sky. And we have to make sure that this is not affected by something mundane like the ability to tell them apart when they're very close to each other. We don't want to count two as one, for example. Or we don't want to be somewhere where there is absorption and the number is diminished because of that. So these are the kind of things that astronomers have to deal with. And they have developed various methods to, you know, check for these things. And our colleague, Nathan, is an expert on these. He's totally meticulous about looking for systematics, which is, you know, again and again, that's what comes to bite you on the ass when you think you have a result. You have to be very, very cautious. But let me just summarize the situation by saying that we have convinced ourselves that what we see is indeed a dipole on the sky and that dipole is twice as large as the dipole in the cosmic microwave background, right? What that means is that normally, since we believe that the cosmic microwave background dipole is the result of our local motion, the idea is that if I do a Lorentz boost at 369 kilometers per second in that frame, I would then see the microwave background as isotropic. The dipole would disappear. Because the dipole is simply because I'm moving with respect to the frame in which everything is isotropic. Right. And the implicit assumption has always been that in that frame, the distribution of matter is also isotropic. Okay, so right now you're distinguishing between a kinematical dipole and a matter dipole? Yeah, so I'm saying the matter dipole should also be kinematical in the standard model—that the dipole that we see in the matter should be exactly due to the same reason that we are seeing a dipole in the radiation, because the two are, well, connected together in the universe, then they decoupled, right? But the last scattering surface of the cosmic microwave background that we see looks pretty isotropic, apart from little fluctuations, right? And then we see this huge dipole, but we say, "Well, that's nothing to do with the early universe; that's to do with just our local motion," right? And by the same token, if we do this boost—we apply this boost to all the things that we measure about matter, namely the redshifts, the luminosity distances—then the belief is, the hope is, that we will have corrected for our local motion and we can then analyze the data according to the Friedmann equations and deduce dark energy, deduce evolving dark energy in the case of DESI, whatever, the usual machinery. But now I'm saying that there is a huge spoke in the wheel. It turns

out that that whole procedure that is followed as standard, including by the DESI collaboration, is now subject to question because we are finding that the dipole in the matter is not the same as the dipole in the CMB. That is the necessary requirement for the standard procedure to go through. Right? So, let me step back one and just describe it in a few short sentences. Let's say the universe is on large scales, homogeneous and isotropic as we want it to be or take it to be in the standard model. In reality, we look out and we see, yes, the cosmic microwave background looks pretty isotropic. That supports what I always thought. Except that there is a huge dipole, a hundred times bigger than the other little fluctuations, which are negligibly small. They are 1 part in 10 to the 5. What is this dipole? And then somebody says, "Oh, no, no, that we understand. That's just due to our local kinematic motion. We just need to correct for it. And then we are back in the reference frame where everything is really isotropic and we can, you know, business as usual. We can use the same model that was invented over a hundred years ago by Alexander Friedmann; that still works." Great. Right. Yeah. And we carry on. That's what DESI are doing. That's what every collaboration is doing. They're using the same equations that were written down by Friedmann and Lemaître nearly a hundred years ago. Okay. I'm saying that to date it has been okay to do that because there was no reason to believe otherwise. But just in the last few years, we have got the data that allows us to do a consistency check of whether the metric of the universe is indeed Friedmann-Lemaître-Robertson-Walker. That is the Ellis-Baldwin test. To do the Ellis-Baldwin test, which is simplicity itself—it's just special relativity—just saying, if you see a dipole in the cosmic background due to a local motion, you should see the same dipole in the sky distribution of cosmologically distant sources. And when we do that test, we find that, yes, there is a dipole in the distribution of distant sources, like quasars and radio sources, but its amplitude does not match. Its amplitude is not what it should be. And that has been now established at more than five standard deviations by multiple data sets: radio data, quasar data, data taken from the ground, data taken from a satellite, data analyzed by three independent groups. We are all getting the same answer. It's significant. So in a recent review in *Reviews of Modern Physics*, we have given a detailed description of these different data sets. We have discussed all the possible systematic uncertainties that people are quite rightly concerned about. And we show that to date, there are no showstoppers. There is no reason to believe that this result is not correct. And if that is the case... Yeah. Okay, so let me see if I got this correct. The critics may say that the evidence for dark energy or Λ is quite overwhelming because there's various sources like supernovae, and then there's CMB, and then there's BAO. And so your response is, well, all of these rely on the FLRW metric, and what warrants the validity of this metric is something called the cosmological principle, which is just this twin of homogeneity and isotropy that you've been referencing. Homogeneity means that if you're at a point P here in this universe and you look out, then it should be spatially the same as if you go to another point, say Q. And then isotropy means that you have rotational invariance. So you're saying that we don't have rotational invariance. Yeah. So the point is that in reality, we accept that the real universe has, of course, got structure. So these statements are reduced to statistical isotropy and statistical homogeneity, which means averaged over a sufficiently large volume, we'll have isotropy and homogeneity. I mean, strictly

speaking, in a universe with fluctuations, you don't ever have exact homogeneity. Even on the scale of the Hubble radius, things are inhomogeneous by about one part in 10 to the 5. But you know, that's small enough that it's sensibly homogeneous, right? We are not going to kind of be too picky about that. However, if that is the case, that the universe was initially smooth and isotropic and homogeneous, then it got little fluctuations, maybe from inflation, whatever—but we don't need those fluctuations because that's what grows into structure. That's what creates galaxies. That's why we are here discussing all this. This structure then means that today's world is inhomogeneous, but the belief is that averaged on sufficiently large scales, we'll still have statistical isotropy and statistical homogeneity. That is the backbone. That is the foundation of today's cosmological model, right? So the cosmological principle that you alluded to, that's a—you know, we shouldn't really do physics by principles, but by empirical evidence. So that was the starting point at a time when there was no data, right? In fact, Weinberg says that in his textbook on gravitation and relativity, he says, "Why do we have this cosmological principle? Well, because there is not much data." But then he says, "When, you know, if the data comes, we really should check this, because nothing could be more interesting than to show that the cosmological principle is wrong." And all I'm saying is that that data has taken a long time to arrive. We have had to wait, you know, till around 2020 to actually have the data. And because of that, now we are in a position to say that the cosmological principle has been falsified at more than five standard deviations because the kinematic dipole in the microwave background does not match what should be the kinematic dipole in the matter distribution. The two are different. And what that then means is that the standard procedure of boosting to this hypothetical so-called CMB frame or cosmic rest frame, as it is sometimes called, in order to apply the equations to the data—which is the standard practice that people do to deduce dark energy, to deduce evolving dark energy in the case of DESI—that entire procedure is now, you know, open to question. And in fact, I'm sufficiently convinced that we have a result that I think we need to go back to square one. We cannot any longer proceed with this Friedmann-Lemaître-Robertson-Walker metric. And there is the rub. You asked earlier, what do I have—what sort of a metric do I prefer? The point is that the FLRW metric is the maximally symmetric metric you can have for space-time. It's a unique simple solution. It's got so much symmetry that you can reduce the 10 coupled equations of Einstein to just one single Friedmann-Lemaître equation and the Raychaudhuri equation for the acceleration, which also simplifies to a simple equation. The point is that only because of that, cosmology became tractable because, you know, everybody can solve this simple differential equation, right? And we do. And then we can confront this simple model with a wide variety of data as it kept coming in. And the data has been, in my view, overtaking the theory for quite some time now. The theory is still that hundred-year-old one. It has been supplemented by a theory of structure formation, which is that you say on top of the homogeneity and isotropy, we have a Gaussian random field of small fluctuations with a scale-invariant spectrum and these grew under gravity. And amazingly, that actually gives a pretty impressive fit to data on a wide variety of scales, right? So that is a success, right? But the basic underlying—under the hood—the basic metric structure is still the FLRW metric from a hundred years ago. The problem is that

once you start dropping symmetry, then Einstein's equations, as you really know quite well, are extremely hard to solve. And at the same time as Einstein gave his solution, there was a parallel solution due to Lemaître, Tolman, and Bondi—LTB—which said, “Well, we’ll still keep isotropy, but maybe we can have inhomogeneity in one direction, in the radial direction; we can have a radial function which is an additional degree of freedom,” right? Incidentally, just that alone is enough to make the supernova evidence go away. If I’m allowed to have a radially varying function which determines how light rays propagate, then I can absorb any difference between distant supernovae and nearby supernovae by suitably adjusting that function. So for example, if you are in a large void or in an over-density or something, but things were still isotropic, that would be enough to do away with the evidence for acceleration. Because you could imagine, as a trivial example, that it could be that locally the Hubble parameter is whatever it is, 70 kilometers per second, let’s say. But let’s say we are in a void, so we are expanding quite fast, and outside the Hubble parameter is 50 kilometers per second, right? And that in itself would be enough to take away all the evidence for acceleration. Wouldn’t an LTB void model require you to be within—to be in the center within, say, 1% because there’s spherical symmetry? Right. You’d have to be pretty close to the center and, you know, then people say that’s fine-tuned. But that’s what I find funny. The same people who think that the LTB model in which we are within a few percent of the center is fine-tuned are prepared to live with a cosmological constant that is fine-tuned not to a few percent, but to one part in 10 to the 60. So that’s what I meant by saying there’s a culture gap, you know, like people don’t realize just how fine-tuned is this cosmological constant that they glibly write down, right? I imagine they would say, well, the cosmological constant, the concordance for it is over-determined because you have multiple independent measurements that agree on this six-parameter model, like you have the CMB for the matter density. But in truth, when you actually look at the data, that ain’t true. First of all, many of those lines—so there are only two or three lines of evidence where the data is actually good enough to say something. Then you have other lines of argument where the data is not actually good enough, but people, when they are fit with the concordance thing, they publish a paper saying, “I looked at, I don’t know, distribution of clusters and this triple correlation function and it is consistent with the standard model.” They don’t tell you when it didn’t fit; they tell you only when it fits. So I know someone who actually did a study of—this is an amusing fact, also to do with Λ —that of the, I don’t remember the precise number, but something like of the 30 measurements of Λ following the WMAP paper which claimed to have established the standard model with Λ , of these only three were outside the one sigma range from the, quote, “true” value of Λ , right? So you know that one sigma is meant to be roughly 67%. So a third of your data points should lie outside one sigma, right? But in their case, only about less than 10% were outside. So there clearly is a selection bias. People, you know, like to be on the winning side. So we really should ignore all the data sets that by themselves don’t have the confidence to claim something; they can only claim consistency with something, which is a different statement. They are consistent with Λ , but they are also consistent with no Λ . So let’s ignore those. Let’s just look at the data sets which definitely say we need Λ , right? Now of these, the supernova data set is the most interesting

because Λ is directly entering into the measurement of the luminosity distance which you observe in the sky, right? When you look at the CMB, you don't actually measure anything to do with Λ . When you look at the CMB, you don't measure Λ . Λ is totally unimportant at the time of CMB decoupling. Because if it is comparable to matter today, the matter density was 10 to the 9 times more important at a redshift of a thousand. Because matter goes as Z^3 . Λ does not change at all. So it was subdominant by a factor of 10 to the 9. What the CMB tells you is that the curvature of space-time—the average value between us and the last scattering surface—that's pretty close to zero, right? It doesn't measure Λ at all. Then you have a measurement of Ω_m from looking at how much matter there is in clusters of galaxies, or the baryon acoustic oscillation scale tells you that. So you have a measurement of Ω_m , you have a measurement of Ω_k , and you have a measurement from the supernovae, which is like a point—I can't remember the exact value, it is something like 0.8 times Ω_m minus—let me get this right. I just have to remind myself. Yeah, it's 0.8 times Ω_m minus 0.6 times Ω_k . That is the combination that is measured by the supernova people, right? And that's slightly negative. So when you put all this together using the sum rule— $\Omega_\Lambda + \Omega_m + \Omega_k = 1$ —then you get Ω_Λ is 0.7. But it is using the sum rule. And the sum rule is directly based on the assumption of homogeneity and isotropy. Because if you had other terms, if you had viscosity, vorticity, angular momentum, whatever, that would be additional terms in the Friedmann equation. So then you don't have just three terms. Yeah. Let me ask you this. In the FLRW, there's an initial singularity. Yes. That's said to be the Big Bang. So if you're discarding the FLRW, are you also contesting the Big Bang model? But the point is the Big Bang is simply a speculation. The FLRW model cannot be extended back to the Big Bang in any case because we know that the whole metric description of space-time itself really is not valid up to the quantum gravity scale. Earlier you said you're going to ask me about inflation. One of the big jokes about the grounds for saying we need inflation is the so-called horizon problem, which says that light could not have traveled far enough to causally connect opposite parts of the sky, which we see as having the same temperature. What people forget is that in order to make that argument, you have to construct a light cone back to T of zero. You have to assume that the FLRW metric holds all the way back to the Big Bang. And we actually don't know if that is the case. That integral that you are calculating may not ever converge if space-time becomes fractal or whatever, you know. So you can't actually, formally speaking, you cannot establish that there is a horizon problem except by assuming that FLRW holds all the way back to the Big Bang, which everybody knows is an outlandish thing to claim, right? What did you mean when you said if space-time becomes fractal? Well, I mean, that is one possibility. And we don't actually have any idea what space-time does at the quantum gravity scale. But it is reasonable to believe that it doesn't stay smooth and exactly isotropic and homogeneous. In fact, it would be extremely unlikely if it was, you know, like the simplest, the most symmetric possible form of space that there is. And yet, for example, the current formulation of string theory assumes that you have Lorentz invariance all the way up to the Planck scale, right? It's an assumption. Ideally speaking, you should be able to compute the background from the propagation of the

strings themselves. And that should be self-consistent. And that was the original proposal. But, you know, it's turned out to be mathematically very difficult. I mean, just even on a fixed background, string theory is extremely complicated. So, you know, I'm not saying it can be done. But strictly speaking, we have no idea of how space-time is going to behave as you approach the Planck scale. And whereas in the standard cosmology, we take it to be still described by the FLRW metric all the way back to the Big Bang, in practice, that makes no sense at all. In fact, the number of particles within a causal horizon decreases to less than one long before you actually get to the Planck scale. So you can't even have a notion of a temperature. You can't have a notion of equilibrium. All these things go out of the window. I mean, not many people know this. You can't have a temperature higher than about 1/100th of the grand unification scale. Above that temperature, there is no interaction that we know which can make particles scatter fast enough to stay in equilibrium. Because the strong interaction is asymptotically free, it becomes weaker and weaker as we go to higher temperatures. Inflation predicts Gaussian statistically isotropic perturbations, though, which is what is observed in the CMB. It does. And that is its greatest appeal, that it actually—if inflation is driven by the vacuum energy of a slowly rolling scalar field, then in that quasi-de Sitter space, you can generate fluctuations in any scalar field. For example, the inflaton, or for that matter in the graviton, you get gravity waves as well. And these fluctuations, because it's such a weakly coupled field, will be very close to Gaussian and it'll have a more or less a scale-invariant spectrum because the thing is rolling on a slope with a very small tilt. So you can get exactly the kind of fluctuations that we know are required to create structure. And this had been pointed out long before inflation by Ed Harrison and Yakov Zeldovich. It's called the Harrison-Zeldovich spectrum, right? We already knew this that we needed this. So then along came this idea that it is due to a slowly rolling scalar field. And that is just what the doctor ordered. So we have grabbed it and we call it inflation, ignoring the fact that that model has got no physical basis whatsoever. I mean, we discussed earlier the cosmological constant problem, right? The cosmological constant problem has been unsolved. We don't know whether vacuum energy couples to gravity at all. And in fact, if it did, then we would not have a universe. But we completely ignored that knowledge that we have when we then start talking about the vacuum energy of a slowly rolling scalar field dominating the universe and driving inflation. And then at some point, somehow it magically falls into its minimum and all the vacuum energy is converted into radiation and we start off the hot Big Bang. So when this BICEP2 experiment claimed to have seen the signature of gravitational waves from inflation, that would have required a cancellation of one part in 10 to the 160 in the energy from the top of the plateau to the bottom of the plateau. That's why I didn't believe it, that it could even be possible. But the point is, for some strange reason, there is this whole industry of inflation. And I have to say, I've written papers on inflation. We are trying to construct models of supersymmetric inflation, which are slightly more predictive. But they all have this fundamental issue that we don't understand how vacuum energy couples to gravity. It's a big, big problem. And to my knowledge, no theory of quantum gravity has addressed that question. It's an unsolved problem. But, you know, to come back to what I was saying, the strict fact is that if we go away from the Friedmann-Lemaître-Robertson-Walker metric, then it is

possible to consider more general metrics. LTB is one, but more generally the so-called Szekeres metrics, which allow for both inhomogeneity and anisotropy. The problem is that the number of parameters proliferate like mad. And we can't pin them down with data, except that some people think that the data that is now going to come from the new missions that are being undertaken today—like this Legacy Survey of Space and Time that will be undertaken by the Rubin Observatory, the SPHEREx, which is a satellite that is currently mapping the sky (it is the successor, in a sense, to the WISE satellite that I mentioned earlier), and then there is Euclid already in orbit taking data and so on—this avalanche of data might be enough that we can actually start fitting more complicated metrics to the data. We can't do that in the old way, but maybe machine learning here will actually be helpful. This is a dream that people have. It might well become reality. But philosophically, the point is this: that sticking to the simple model means that more people can play. It means that astronomers, radio astronomers, particle theorists, everybody can do cosmology because the maths is so simple, right? If the maths becomes complicated—very complicated—then I'm afraid it will become a more arcane activity. It will not be accessible. It will not be as inclusive as it is today. And what this really reminds me of—I was complaining to a senior astrophysicist that we have such hard evidence and that people are not taking much notice of it. And he said, “You know, did you know that the evidence for continental drift, it took 50-plus years for it to be accepted, even though the evidence was overwhelming?” It's not just that the continents seem to fit together. There was fossil evidence, right? And this was already known in the 1920s, '30s, right? But it took 50 years before it was accepted because there was no theory of tectonic plates. People didn't know, couldn't figure out how could the continents move about. So they needed to have a physical understanding of what this evidence was telling us before they accepted the evidence, right? And in a strange way, I think that is being repeated now. We have evidence, but it will not be accepted until we provide an alternative theoretical background for how we could have a mismatch between the matter frame and the CMB frame. And then people can start fitting the data to that new model. That's what they want to do, right? Interesting. Now do you think your alternate model—that is, there's question marks over right now because you don't know what it is—do you think it will ultimately be another metric or do you think it'll be a theory of quantum gravity? Well, the cosmological constant problem certainly has not gone away. And even if I say that dark energy does not exist, that doesn't solve the cosmological constant problem. That problem is still there. The cosmological constant problem is overwhelming. The problem is: why are we here at all? How is it that the universe can possibly have expanded to this huge size and be expanding at only, you know, 70 kilometers per second per megaparsec after whatever 14 billion years, when its natural time should have been—A, it should have been set by the time at which we reached the energy scale at which the standard model vacuum energy would have kicked in. That would have been something like 10 to the minus 12, 13 seconds after the Big Bang, right? When the universe was no more than a millimeter big. That is how big we could have got before vacuum energy became the most important thing in town and took over, right? It didn't. We are here to tell the tale. And yet, we seem to have forgotten that. And today, we start inventing models of the early universe where we invoke vacuum energy and then lose

it. As the famous physicists used to say, “Use it, then lose it.” That’s what you’re doing. That’s inflation, right? So, maybe there is a clue there. Maybe inflation lasted for a short time. Some people are thinking today that de Sitter space itself is something which is abhorrent to nature. You know, you can’t have de Sitter space because in de Sitter space, there is no future infinity. When you scatter two particles, you can’t check unitarity. You can’t check the outgoing states at future infinity because there is no such thing. And that is one reason to believe. This was apparent from the very early days of string theory because this is what was realized back in 1983, when the first anomaly cancellation theorems came up, etc. So people do worry about it. And actually it is that realization that has led to this notion that you would have—you know about this notion of this swampland and landscape of string theory, right? That is all driven by this realization that there is something fundamentally dodgy about de Sitter space, right? You can’t have de Sitter space in nature. You have to have an arrow of time. You have to have a little tilt to it, right? And then, of course, you have to worry about the cosmological constant problem itself, right? So, basically, the cosmological constant—vacuum energy of whatever sign—is a huge problem in our most fundamental theories. In string theory, as you know, you typically get a negative cosmological constant; you get anti-de Sitter space. And then they have to do a bit more manipulation to try to make it slightly de Sitter because that’s what they think the real world is like. Why? Because the astronomers have told them so, right? Even if I tell them that, “Okay, it’s not de Sitter, it is zero,” even that would involve the same fine-tuning. So, you know, as far as they’re concerned, it might as well be hung for a sheep as for a lamb, you know. It’s a very difficult situation. And on the one hand it’s intensely frustrating that you have made no headway on that problem. On the other hand, I think it is actually pretty encouraging because it means that we really—there is something very big we are yet to find. This is—I mean, I personally may not be able to kind of make any progress towards that. I mean, certainly brighter people than me have failed, if that’s any comfort. But I am always reminded of something that David Gross—I heard him say. He says, “My dog doesn’t understand quantum mechanics. You know, my dog is pretty smart. It brings me in the newspaper. When I talk to it, it listens like it understands what I’m saying. But if I say ‘Schrödinger’s equation’ to it, nada. No response. It definitely does not understand quantum mechanics.” So maybe there are some things that we’ll never understand, you know, just we won’t be able to solve that problem. Now that may be true, but it is also true that we are never going to stop trying. We are not going to ever say, “We have found a problem; it’s uncrackable; we’re giving up; we are going off home,” right? So I think we are not going to make progress in cosmology, I think, until we solve the cosmological constant problem, because everything else will be just patching stuff up. But certainly, if there is empirical evidence that the universe cannot be described by the Friedmann-Lemaître-Robertson-Walker metric, as I think we have established, then it behooves us to find a better description geometrically. And Ellis, George Ellis, had already given a prescription for how to do that. He called it the cosmological fitting problem, also 40 years ago, right? And he talked about how to actually use observations to construct the metric of the universe, not start with an assumption and try to see if it fits the data, but use the data to infer the metric from. And that is, again, a program that would have been inconceivable some years ago, but now might be possible to do if you

can frame it in the right way, using all these new tools that are available of dealing with huge masses of data, huge number of parameters, you know, all these advances in machine learning, etc., maybe you should do it. Now, I'm not saying that we will do it. I'm saying that there is a large community of cosmologists out there and most of them are busy doing parameter fitting. And it would be great if at least some of them join us in trying to find this new description. If you like, this equivalent of the theory of tectonic plates, that will then make everybody accept that, yes, the continents are moving. And, you know, we have a physical framework for how to understand that. Professor, what is a piece of advice that you found to be true, that you come back to often, that was told to you or given to you by a senior, by one of your professors, by a colleague? Well, I actually started out my research career doing experimental cosmic ray physics. And it was actually after my PhD when I first went to Oxford that I met this very charismatic relativist, Dennis Sciama, who I mentioned earlier, who had already, you know, he had mentored many famous people. And Dennis had written a book back in the 1970s, 1971, I think, called *The Unity of the Universe*. And he wrote very well. And I remember being struck by the fact that Dennis was a mathematical relativist. He did rigorous work and also on quantum field theory in curved space-time and so on. But he wrote, "None of us can understand why there is a universe at all. Why anything should exist? That is the ultimate question. But while we cannot answer this question, we can make progress with the next simpler one: what the universe as a whole is actually like." Right. In other words, here was Dennis telling us that our first instinct when we come to cosmology is to ask the deep questions, the philosophical questions, right? What is this all about? Why we are here? What does it all mean? What is our place in the universe? But he says, "Be empirical. Be like a physicist. First describe the universe," right? And in a way, that is what we are now doing. I mean, you know, it might seem a bit late, but actually I'm saying that our description of the universe for the past century—which was the sensible thing to do at the time, describe it as the isotropic homogeneous substrate, right?—on closer examination, as more data comes in, it turns out that that is not quite true. We are not talking about a huge anisotropy. We are talking about 0.5 of a percent. It's tiny, right? But it is important enough that it gets rid of two-thirds of the energy density of the universe, which is supposed to be in the form of a cosmological constant or dark energy. That's how important this anisotropy is. Even though, strictly speaking, you would not be able to see it on the sky unless you had statistics of millions of objects. It's so small. Even in the cosmic microwave background, you see this anisotropy of one part in a thousand because there are billions of photons. So you can see the pattern quite clearly. With astronomical objects, you have to work a lot harder. It's a tiny effect. But it undermines our notion of the perfect symmetry that our model assumed. Now that should not surprise us. In every other place in physics we have come across the idea that perfect symmetries are broken or hidden in the real universe we live in, right? There was a symmetry between weak and electromagnetic interactions in the early universe. Today, there isn't. That is why the W boson has a mass and the photon doesn't, right? That is a major effect, that breaking up that symmetry. And this notion of spontaneously broken or spontaneously hidden symmetries is fundamental to our construction of the standard model of gauge forces, right? And yet in cosmology, we have held on to this notion of the, you know, what

graduate students call the spherical cow approximation for over a century. I mean, so, you know, when I say it, people often say, “So what’s so great about that? But after all, I’ve never believed that the universe should be perfectly spherical or isotropic or whatever.” And I say, “Yeah, well, that is true. But did you know that the inference that Ω_{Λ} is 0.7—in other words, dark energy makes up two-thirds of the energy of the universe—that that is based on making exactly that assumption?” That connection has not been made by many people. So a lot is at stake because our deduction that we live in a universe dominated by dark energy, which is accelerating, is ultimately based on this very strong assumption of isotropy and homogeneity. And that is what is called into question by the data tests that we have and others have performed. So I think in that respect, it is not particularly complicated or deep. It’s very simple and that is why I think for the same reason it should be more easy for people to understand what this is about and to take it on board. And, you know, we could talk more about the philosophical implications of this, but maybe another time. Yeah. Professor, thank you so much for speaking with me for so long. And I know you’re staying up past midnight and I know we’re both dealing with sleep issues of various sorts. So I’m glad that we’re able to find the time. Well, thank you for—yeah, I’m afraid, as you can see, I’m actually quite engaged with this subject. I’ve been working on it now for quite some time. And it has been frustrating to not have ready acceptance of what we find and a lot of pushback. And I think that’s part of the scientific method. But I do hope that the younger generation of cosmologists will start thinking with a more of an open mind than has been the case for the past decade or two when this so-called standard model of cosmology has got such a grip on people’s minds that they think that there is no other game in town. And this, I think, is completely against the spirit in which science should be done and especially cosmology should be done. So I hope that that change has now started happening. And I thank you for giving me a platform to say this because I know a lot of people who may not have read our paper in detail might listen to this and think, “Well, maybe I should take a look.” And that would be reward enough for me. Hi there, Curt here. If you’d like more content from Theories of Everything and the very best listening experience, then be sure to check out my Substack at CurtJaimungal.org. Some of the top perks are that every week you get brand new episodes ahead of time. You also get bonus written content exclusively for our members. You can also just search my name and the word Substack on Google. Since I started that Substack, it somehow already became number two in the science category. Now, Substack, for those who are unfamiliar, is like a newsletter, one that’s beautifully formatted. There’s zero spam. This is the best place to follow the content of this channel that isn’t anywhere else. It’s not on YouTube. It’s not on Patreon. It’s exclusive to the Substack. It’s free. There are ways for you to support me on Substack if you want, and you’ll get special bonuses if you do. Several people ask me, like, “Hey, Curt, you’ve spoken to so many people in the field of theoretical physics, of philosophy, of consciousness. What are your thoughts, man?” Well, while I remain impartial in interviews, this Substack is a way to peer into my present deliberations on these topics. And it’s the perfect way to support me directly. CurtJaimungal.org or search Curt Jaimungal Substack on Google. Oh, and I received several messages, emails, and comments from professors and researchers saying that they recommend Theories of Everything to their students. That’s fantastic. If

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